In some real world problems, we can often easily find some rate of change with a different unknown rate being related some how. For example, filling a balloon, It's easy to measure the change in volume (how much air is added) but we don't have a direct measurement of the change in radius (or width); since radius and volume are directly related, we can find the rate of change in radius.

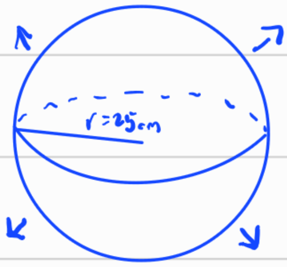
We'll spend this section going through various examples to get a grasp of the process.

As a rule of thumb <u>ALWAYS</u> draw a picture.

Ex: Air is being pumped into a spherical balloon so it's volume is increasing by $100\,cm^3/s$. How fast is radius changing when diameter is 50 cm?

Given: $\frac{dV}{dt} = 100\,cm^3/s$, Unknown: $\left.\frac{dr}{dt}\right|_{r=25cm}$

Useful formula: $V = \frac{4}{3}\pi r^3$



Now, we treat V and r as functions of time, this is reasonable as they do change over time. We can then differentiate the entire equation with respect to time.

$$\frac{d}{dt}(V) = \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) \longrightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

we want to find $\left.\frac{dr}{dt}\right|_{r=25}$, so

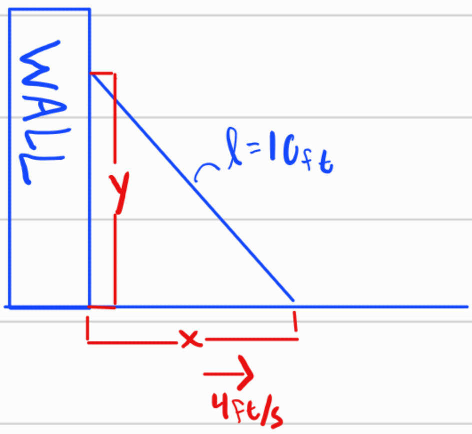
we know this from earlier

$$\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt} \longrightarrow \left.\frac{dr}{dt}\right|_{r=25} = \frac{1}{4\pi(25)^2}\frac{dV}{dt}$$

$$= \frac{1}{4\pi(25)^2} 100$$

$$\boxed{= \frac{1}{25\pi} \text{ cm/s}}$$

Ex: A ladder (10ft long) rests against a vertical wall. If the base of the ladder slides away from the wall at a rate of 4 ft/s how fast is the top of the ladder sliding down the wall the ladder is 6ft from the wall?



Known $l = 10$, $\frac{dx}{dt} = 4\,\text{ft/s}$

Want $\left.\frac{dy}{dt}\right|_{x=6}$

- Find relation of x and y:

From Pythagorean Thm, $x^2 + y^2 = 10^2 = 100$

- Note, if $x = 6$, $y = \sqrt{100-36} = 8$

- Differentiate

$$\frac{d}{dt}(x^2+y^2) = \frac{d}{dt}(100)$$

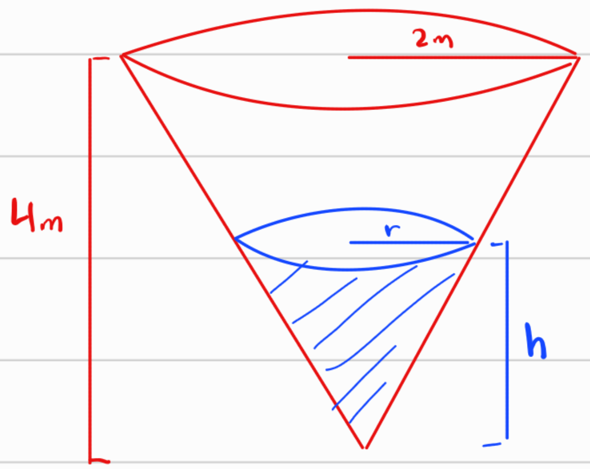
$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{x}{y}\underbrace{\frac{dx}{dt}}_{\text{known}} = -4\frac{x}{y}$$

- Plug in Values

$$\left.\frac{dy}{dt}\right|_{\substack{x=6\\y=6}} = -\frac{6}{8}\cdot 4 = \boxed{-3\,\text{ft/s}}$$

Ex: A water tank is shaped like an inverted cone with a base radius 2m and height 4m.
If water is being pumped in at a rate of $2m^3/min$, find the rate at which the water level is rising when it is 3m deep.



Known: $\frac{dV}{dt} = 2m^3/min$

want $\left.\frac{dh}{dt}\right|_{h=3}$

Volume of Cone is $V = \frac{1}{3}\pi r^2 h$.
We need to eliminate r as we don't know or need $\frac{dr}{dt}$.
In a cone, all smaller cones are similar so $\frac{r}{h} = \frac{2}{4} \to \frac{h}{2} = r$
So Volume, in this case, is

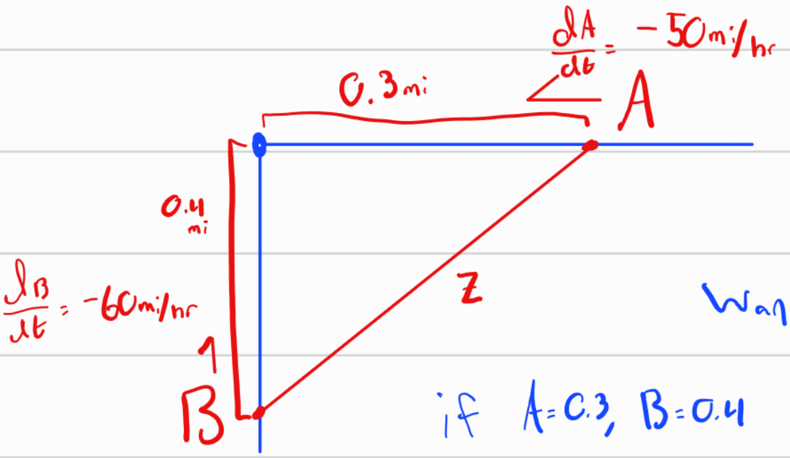
$$V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi}{12}h^3$$

then $$\frac{dV}{dt} = \frac{\pi}{4}h^2\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{4}{\pi h^2}\overbrace{\frac{dV}{dt}}^{\text{Known, } = 2m^3/min} = \frac{8}{\pi h^2},$$

so $$\left.\frac{dh}{dt}\right|_{h=3} = \boxed{\frac{8}{9\pi} m/min}$$

Ex: Car A is traveling West at 50 mi/h and Car B is heading North at 60 mi/hr. Both Cars are heading towards an intersection. At what rate are the two cars approaching each other when Car A is 0.3 mi and car B is 0.4 mi from the intersection.



Want $\left.\frac{dz}{dt}\right|_{z=0.5}$

if $A = 0.3$, $B = 0.4$

then $z = 0.5$

Find eqn

$$A^2 + B^2 = z^2$$

Pythagorean Thm

Differentiate

$$2A\frac{dA}{dt} + 2B\frac{dB}{dt} = 2z\frac{dz}{dt}$$

rewrite

$$\frac{A\frac{dA}{dt} + B\frac{dB}{dt}}{z} = \frac{dz}{dt}$$

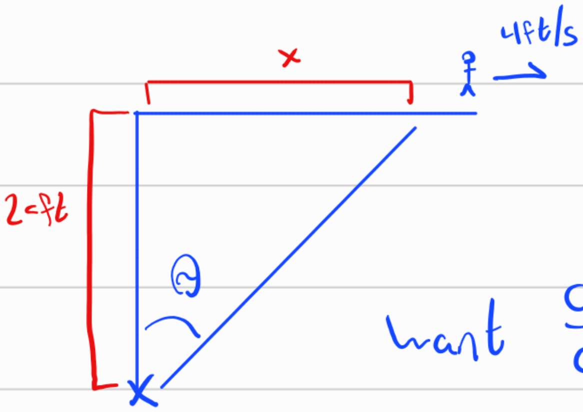
Solve

$$\frac{0.3(-50) + 0.4(-60)}{0.5} = \frac{dz}{dt}$$

$$2(-15 - 24) = -78 = \frac{dz}{dt}$$

$$\boxed{\frac{dz}{dt} = -78 \text{ mi/hr}}$$

Ex: A man is walking at 4ft/sec on a Strait Path. 20 ft Perpendicular to the Path is a spotlight following the man. At what rate is the light turning when the man is 15ft from the Point on Path closest to the light.



Know $\frac{dx}{dt} = -4\text{ft/s}$

want $\left.\frac{d\theta}{dt}\right|_{x=15}$

Find eqn

$$\tan(\theta) = \frac{x}{20}$$

Differentiate

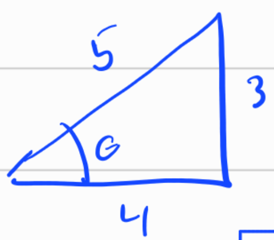
$$\sec^2(\theta)\frac{d\theta}{dt} = \frac{1}{20}\frac{dx}{dt}$$

Rewrite

$$\frac{d\theta}{dt} = \frac{\cos^2(\theta)}{20}(4)$$

To find $\cos(\theta)$, when $x = 15$, we have $\tan(\theta) = \frac{15}{20} = \frac{3}{4}$

so



$\rightarrow \cos(\theta) = \frac{4}{5}$

$$\left.\frac{d\theta}{dt}\right|_{x=15} = \frac{(4/5)^2}{20}(4) = \boxed{\frac{16}{125}\text{ rads/sec}}$$